Some Fourier transform theory

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1 Definitions of the Fourier transform:

Definition 1.1. *Let $f : \mathbb{R}^n \to \mathbb{C}$, $f \in L^1(\mathbb{R}^n)$, and λ denote the Lebesgue measure on $\mathbb{R}^n$ then we have the Fourier transform of f:*

$$\mathcal{F}\{f(x)\}(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}d\lambda(x).$$

It is evident why the Fourier transform is well defined as a consequence of $f \in L^1(\mathbb{R}^n)$:

$$\left|\int_{\mathbb{R}^n} f(x)e^{-2\pi i\xi\cdot x}dx\right| \leq \int_{\mathbb{R}^n} |f(x)|\,dx, (|e^{-2\pi i\xi\cdot x}| = 1).$$

Of course, this is not the only convention for the Fourier transform:

Definition 1.2.

$$\hat{f}(\omega) = \int_{\mathbb{R}^n} f(x)e^{-i\omega\cdot x}d\lambda(x),$$

$$\hat{f}(\omega) = \frac{1}{\sqrt{2\pi}^n}\int_{\mathbb{R}^n} f(x)e^{-i\omega\cdot x}d\lambda(x).$$

$$\hat{\psi}(p,t) = \frac{1}{\sqrt{2\pi\hbar}^n}\int_{\mathbb{R}^n} \psi(x,t)e^{-\frac{ip\cdot x}{\hbar}}d\lambda(x), (p = \hbar k)$$

The last one is actual the momentum-space Fourier transform, which is imperative in Quantum theory (though it was most probably made obvious by the presence of the Planck constant.)

2 Certain properties of the Fourier transform:

Theorem 2.1 (The scaling property of the Fourier transform)**.** *For a function $f \in L^1(\mathbb{R}^n)$ and $a \in \mathbb{R} \setminus \{0\}$ we have:*

$$\mathcal{F}\{f(ax)\}(\xi) = \frac{1}{a^n}\hat{f}\left(\frac{\xi}{a}\right)$$

Proof. Let $f \in L^1(\mathbb{R}^n)$ and $a \in \mathbb{R} \setminus \{0\}$, we must first prove that the function $f(ax) \in L^1(\mathbb{R}^n)$:

$$\|f(ax)\|_1 = \int_{\mathbb{R}^n} |f(ax)| d\lambda(x) = \int_{\mathbb{R}^n} |f(x)| \cdot \frac{1}{a^n} d\lambda(x) = \frac{1}{a^n}\|f\|_1 < \infty \implies f(ax) \in L^1(\mathbb{R}^n).$$

$$\mathcal{F}\{f(ax)\}(\xi) = \int_{\mathbb{R}^\ltimes} f(ax) e^{-2\pi i \xi \cdot x} d\lambda(x) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot \left(\frac{x}{a}\right)} \cdot \frac{1}{a^n} d\lambda(x)$$
$$= \frac{1}{a^n} \int_{\mathbb{R}^n} f(x) e^{-2\pi i \left(\frac{\xi}{a}\right) \cdot x} d\lambda(x) = \frac{1}{a^n} \hat{f}\left(\frac{\xi}{a}\right)$$

□

Theorem 2.2. *A subset of the set of eigenfunctions of the Fourier transform over $\mathbb{R}^n$ is the set:*

$$\{e^{-a\|x\|^2} \big| \Re(a) > 0\}$$

.

Proof. Let us first compute the Fourier transform of $e^{-\pi\|x\|^2}$ (assuming $x = (x_1, \ldots, x_n)$ and $\xi = (\xi_1, \ldots, \xi_n)$):

$$\mathcal{F}\{e^{-\pi\|x\|^2}\}(\xi) = \int_{\mathbb{R}^n} e^{-\pi\|x\|^2} e^{-2\pi i \xi \cdot x} d\lambda(x) = \int_{\mathbb{R}^n} e^{-\pi\|x\|^2 - 2\pi i \xi \cdot x} d\lambda(x)$$

$$= \int_{\mathbb{R}^n} e^{-\pi \sum_{k=1}^n x_k^2 - 2\pi i \sum_{k=1}^n \xi_k x_k} d\lambda(x_1, \ldots, x_n) = \int_{\mathbb{R}^n} e^{-\pi \cdot \sum_{k=1}^n \left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_1, \ldots, x_n)$$

$$= \int_{\mathbb{R}^n} \prod_{k=1}^n e^{-\pi\left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_1, \ldots, x_n) = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi\left(x_k^2 - 2i\xi_k x_k\right)} d\lambda(x_k)$$

$$= \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi(x_k^2 - 2i\xi_k x_k + (i\xi_k)^2 - (i\xi_k)^2)} d\lambda(x_k) = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} e^{-\pi(-i\xi_k)^2} d\lambda(x_k)$$

$$= \prod_{k=1}^n e^{-\pi\xi_k^2} \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} d\lambda(x_k) = e^{-\pi \sum_{k=1}^n \xi_k^2} \int_{-\infty}^{\infty} e^{-\pi(x_k - i\xi_k)^2} d\lambda(x_k)$$

$$= e^{-\pi\|\xi\|^2} \prod_{k=1}^n \int_{-\infty+i\xi_k}^{\infty+ik} e^{-\pi x_k^2} d\lambda(x_k).$$

We have by Cauchy's integral theorem that $\int_{-\infty+i\xi_k}^{\infty+i\xi_k} e^{-\pi x_k^2} d\lambda(x_k) = \int_{-\infty}^{\infty} e^{-\pi x_k^2} d\lambda(x_k) = 1$

$$\implies \mathcal{F}\{e^{-\pi\|x\|^2}\}(\xi) = e^{-\pi\|\xi_k\|^2}$$

And by the scaling property:

$$\mathcal{F}\{e^{-\pi a\|x\|^2}\}(\xi) = \frac{1}{\sqrt{a}^n} e^{-\pi \frac{\|\xi\|^2}{a}}.$$

□

3 The Fourier inversion theorem:

Naturally, we have to discuss inversion of the Fourier transform, allowing us to obtain a function from a Fourier transform, an imperative step in solving many differential equations with the Fourier transform, which leads us to the next topic:

Theorem 3.1. *Given $f \in L^1(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$, and assuming the first convention, we have:*

$$\int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} dx = f(x).$$

Proof. Let $f \in L^1(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$.

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot x} d\lambda(x)$$

$$\implies \hat{f}(\xi) e^{-2\pi i \xi \cdot y} = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot (y-x)} d\lambda(x).$$

We use Gaussian regularization using the function $e^{-\pi \epsilon^2 |\xi|^2}$:

$$\hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi \epsilon^2 |\xi|^2} = \int_{\mathbb{R}^n} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi \epsilon^2 \|\xi\|^2} d\lambda(x)$$

$$\implies \int_{\mathbb{R}^k} \hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi \epsilon^2 \|\xi\|} d\lambda(\xi) = \int_{\mathbb{R}^n} \int_{\mathbb{R}^k} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi \epsilon^2 \|\xi\|^2} d\lambda(x) d\lambda(\xi)$$

by Fubini's theorem:

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^k} f(x) e^{-2\pi i \xi \cdot (y-x)} e^{-\pi \epsilon^2 \|\xi\|^2} d\lambda(x) d\lambda(\xi) = \int_{\mathbb{R}^n} f(x) \left(\int_{\mathbb{R}^n} e^{-\pi \epsilon^2 \|\xi\|^2} e^{-2\pi i \xi \cdot (y-x)} d\lambda(\xi) \right) d\lambda(x)$$

$$= \int_{\mathbb{R}^n} f(x) \frac{1}{\epsilon^n} e^{-\pi \frac{\|y-x\|^2}{\epsilon^2}} d\lambda(x)$$

Using the substitution:

$$x = \Phi(u) = y - \epsilon u \implies \det(J_\Phi) = \epsilon^n$$

$$\implies \int_{\mathbb{R}^n} f(x) \frac{1}{\epsilon^n} e^{-\pi \frac{\|y-x\|^2}{\epsilon^2}} d\lambda(x) = \int_{\mathbb{R}^n} f(y - \epsilon u) \frac{1}{\epsilon^n} e^{-\pi \|u\|^2} \epsilon^n d\lambda(u) = \int_{\mathbb{R}^n} f(y - \epsilon u) e^{-\pi \|u\|^2} d\lambda(u).$$

$$|f(y - \epsilon u)| \leq \|f\|_\infty = \sup_{x \in \mathbb{R}^n} |f(x)| \implies \left| f(y - \epsilon u) e^{-\pi \|u\|^2} \right| \leq \|f\|_\infty e^{-\pi \|u\|} \in L^1(\mathbb{R}^n)$$

by the dominated convergence theorem:

$$\lim_{\epsilon \to 0^+} \int_{\mathbb{R}^n} f(y - \epsilon u) e^{-\pi \|u\|^2} d\lambda(u) = \int_{\mathbb{R}^n} \lim_{\epsilon \to 0^+} f(y - \epsilon u) e^{-\pi \|u\|^2} d\lambda(u) = f(y) \int_{\mathbb{R}^n} e^{-\pi \|u\|^2} d\lambda(u) = f(y)$$

And since $\hat{f} \in L^1(\mathbb{R}^n)$, we have:

$$\left| \hat{f}(\xi) e^{-2\pi i \xi \cdot y} e^{-\pi \epsilon^2 \|\xi\|} \right| \leq \left| \hat{f}(\xi) \right| \in L^1(\mathbb{R}^n).$$

By the dominated convergence theorem:

$$\lim_{\epsilon\to 0^+}\int_{\mathbb{R}^n}\hat{f}(\xi)e^{-\pi\epsilon^2\|\xi\|^2}e^{-2\pi i\xi\cdot y}d\lambda(\xi)=\int_{\mathbb{R}^n}\lim_{\epsilon\to 0^+}\hat{f}(\xi)e^{-\pi\epsilon^2\|\xi\|^2}e^{-2\pi i\xi\cdot x}d\lambda(\xi)=\int_{\mathbb{R}^n}\hat{f}(\xi)e^{-2\pi i\xi\cdot y}d\lambda(y)$$

$$\therefore f(y)=\int_{\mathbb{R}^n}\hat{f}(\xi)e^{-2\pi i\xi\cdot y}d\lambda(y).$$

Thus, the proof is concluded. □

4 The Fourier transform and the L^2 inner product, Plancherel's theorem.

Definition 4.1 (The L^2 inner product)**.** *Let $(X,\mathcal{A},\mu)$ be a measure space. The Mathematics convention for the L^2 inner product is defined as follows:*

$$\forall f,g\in L^2(X,\mu):\langle f,g\rangle_{L^2}=\int_X f(x)\overline{g(x)}d\mu(x)$$

and:

$$\forall f\in L^2(X,\mu):\|f\|_2^2=\langle f,f\rangle$$

The Physics convention for the L^2 inner product is defined as follows:

$$\forall\psi,\varphi\in L^2(X,\mu):\langle\psi|\phi\rangle=\int_X\overline{\psi}(x)\phi(x)d\mu(x)$$

and:

$$\forall\psi\in L^2(X,\mu):\|\psi\|_2^2=\langle\psi|\psi\rangle.$$

We will proceed with the Mathematics convention, the Physics convention does not change much.

Theorem 4.1 (Self-adjoint-ness of the Fourier transform)**.** *The Fourier transform operator $\mathcal{F}$ is self-adjoint in the L^2 inner product, that is:*

$$\forall f,g\in L^1(\mathbb{R}^n)\bigcap L^2(\mathbb{R}^n):\langle\hat{f},g\rangle_{L^2}=\langle f,\hat{g}\rangle_{L^2}.$$

Proof. Let $f,g\in L^1(\mathbb{R}^n)\bigcap L^2(\mathbb{R}^n)$ with $\hat{f}\in L^1(\mathbb{R}^n)\wedge\hat{g}\in L^1(\mathbb{R}^n)$ (to allow using The Fourier inversion theorem.)

$$\langle\hat{f},g\rangle_{L^2}=\int_{\mathbb{R}^n}\hat{f}(\xi)\overline{g(\xi)}d\lambda(\xi)=\int_{\mathbb{R}^\ltimes}\hat{f}(\xi)\overline{\left(\int_{\mathbb{R}^n}\hat{g}(x)e^{-2\pi i\xi\cdot x}d\lambda(x)\right)}d\lambda(\xi)$$

$$=\int_{\mathbb{R}^n}\hat{f}(\xi)\left(\int_{\mathbb{R}^n}\overline{\hat{g}(x)}e^{2\pi i\xi\cdot x}d\lambda(x)\right)d\lambda(\xi)=\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\hat{f}(\xi)\overline{\hat{g}(x)}e^{2\pi i\xi\cdot x}d\lambda(x)d\lambda(\xi).$$

By Fubini's theorem:

$$\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\hat{f}(\xi)\overline{\hat{g}(x)}e^{2\pi i\xi\cdot x}d\lambda(x)d\lambda(\xi)=\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\hat{f}(\xi)\overline{\hat{g}(x)}e^{2\pi i\xi\cdot x}d\lambda(\xi)d\lambda(x)$$

$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\lambda(\xi) \right) \overline{\hat{g}(x)} d\lambda(x) = \int_{\mathbb{R}^n} f(x) \overline{\hat{g}(x)} d\lambda(x) = \langle f, \hat{g} \rangle_{L^2}.$$

Thus, the proof is concluded. □

Theorem 4.2 (Plancherel's theorem/Rayleigh's theorem.)**.** *For any function* $f \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ *with* $\hat{f} \in L^1(\mathbb{R}^n)$

Proof. Let $f \in L^1(\mathbb{R}^n) \bigcap L^2(\mathbb{R}^n)$ with $\hat{f} \in L^1(\mathbb{R}^n)$.

By Theorem 4.1:

$$\|\hat{f}\|_2^2 = \langle \hat{f}, \hat{f} \rangle_{L^2} = \langle f, f \rangle_{L^2} = \|f\|_2^2$$
$$\implies \|f\|_2^2 = \|\hat{f}\|_2^2$$

Thus, the proof is concluded. □